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Lab Report: ARP Poisoning (Man-in-the-Middle) using Ettercap

Objective

The objective of this lab is to demonstrate the setup and execution of an ARP poisoning Man-in-the-Middle (MITM) attack using the Ettercap graphical user interface (GUI) on a Kali Linux environment.

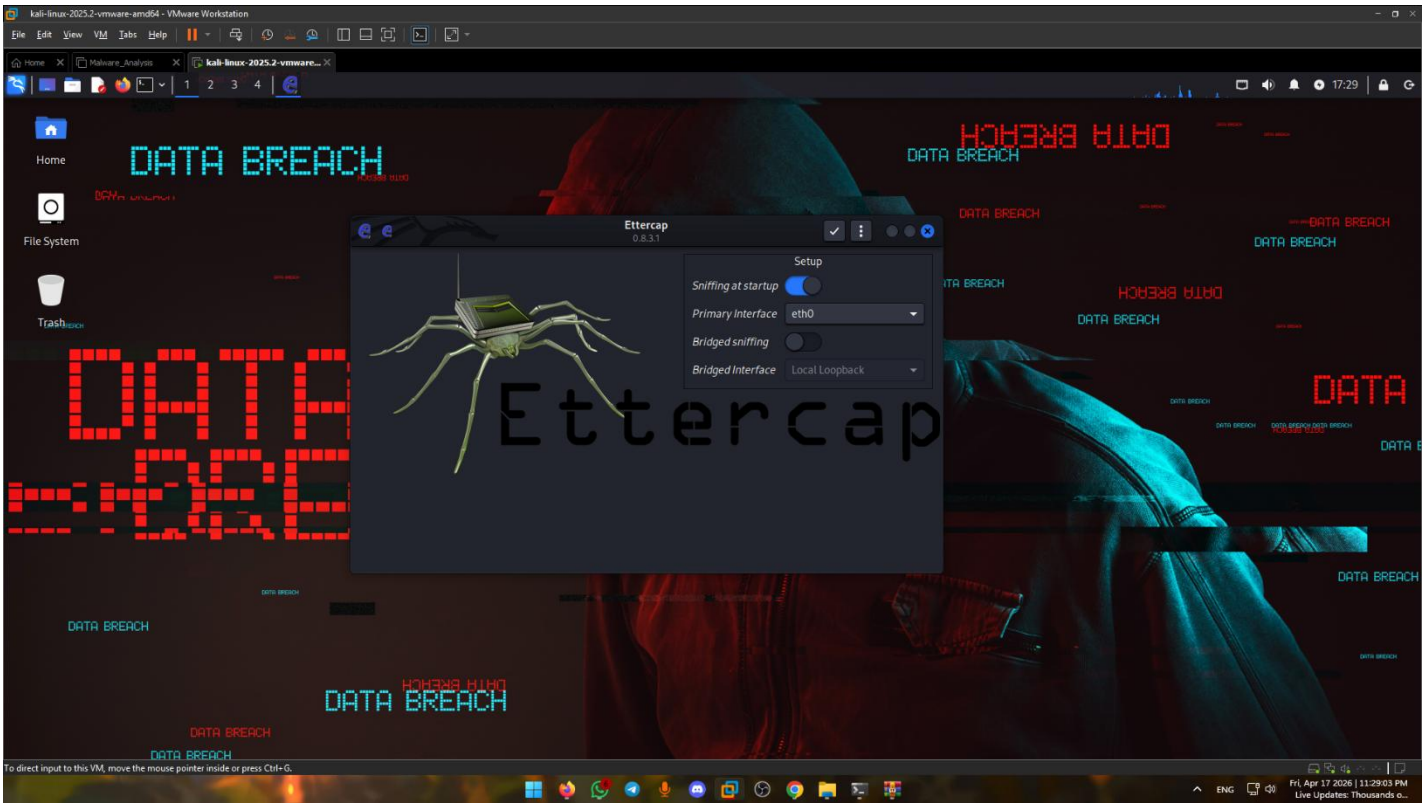
Environment & Tools

- **Operating System:** Kali Linux
- **Tool Used:** Ettercap (GUI version)
- **Primary Network Interface:** eth0

Procedure

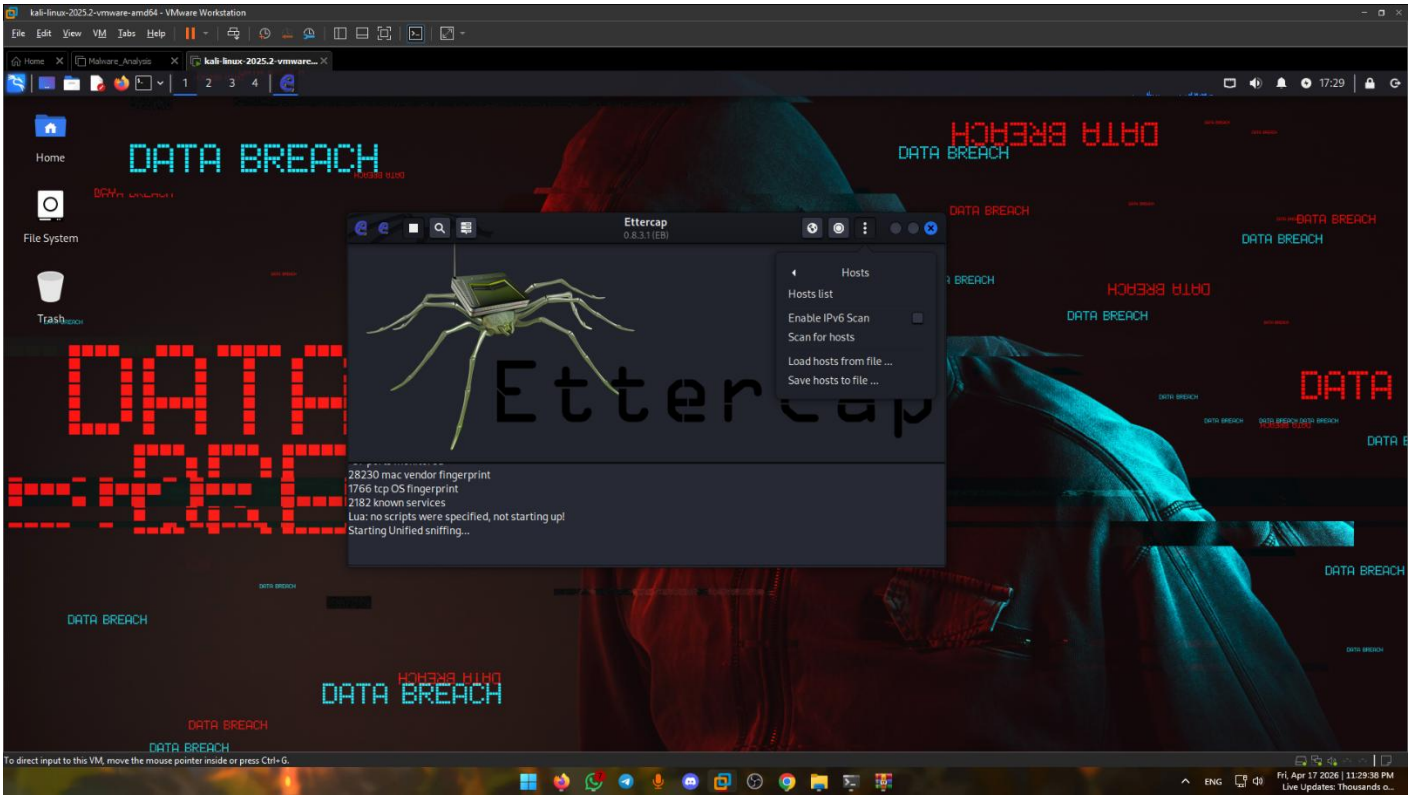
Step 1: Network Interface Identification

- A terminal was opened, and the ifconfig command was executed to identify the attacker machine's network details.
- The output confirmed the primary interface as eth0 with the IPv4 address 192.168.10.128 and a netmask of 255.255.255.0.



Step 2: Launching and Configuring Ettercap

- Ettercap was located via the Kali Linux application menu and launched with root privileges (sudo ettercap -G).
- In the initial setup screen, the "Sniffing at startup" option was enabled, and eth0 was selected from the dropdown menu as the primary interface for packet sniffing.



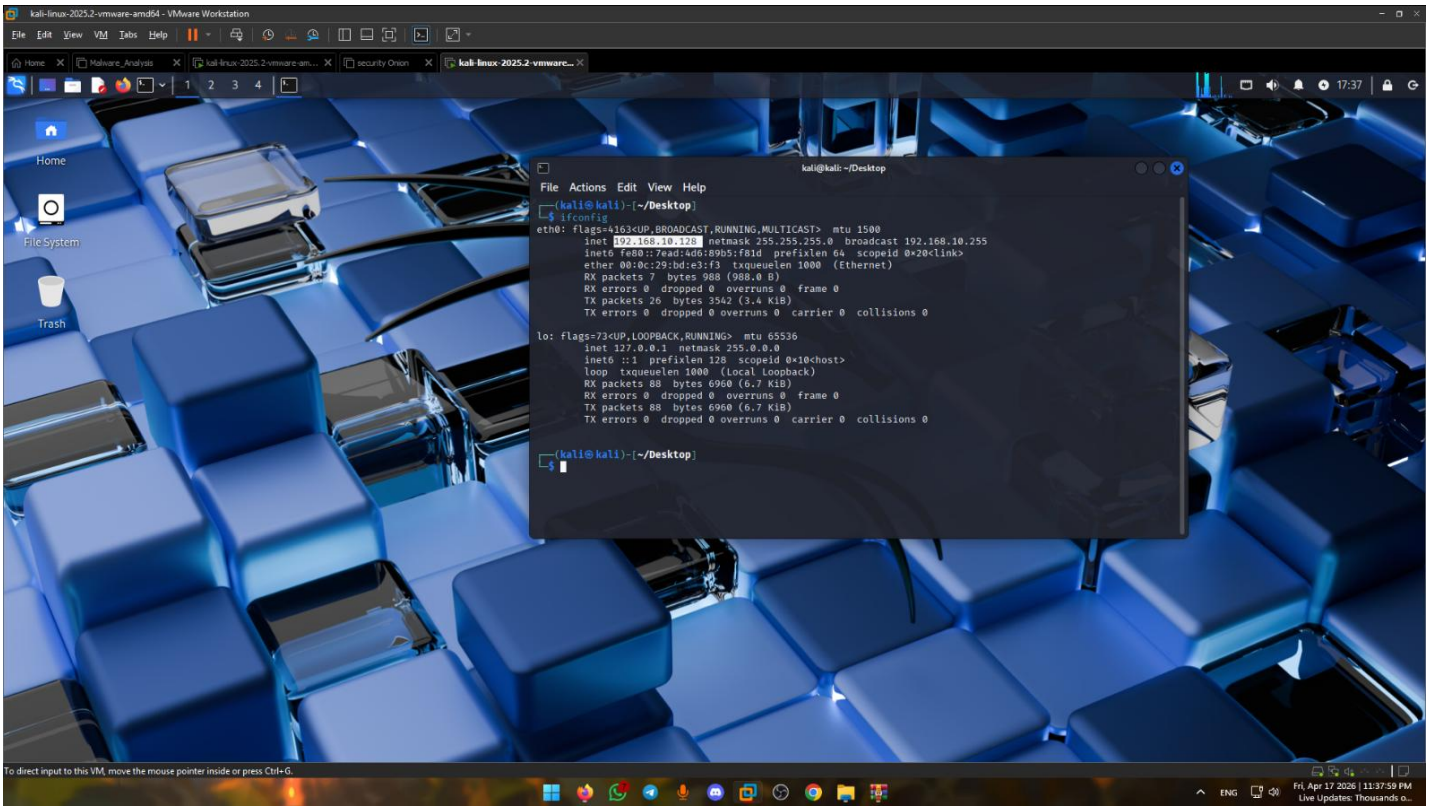
Step 3: Host Discovery

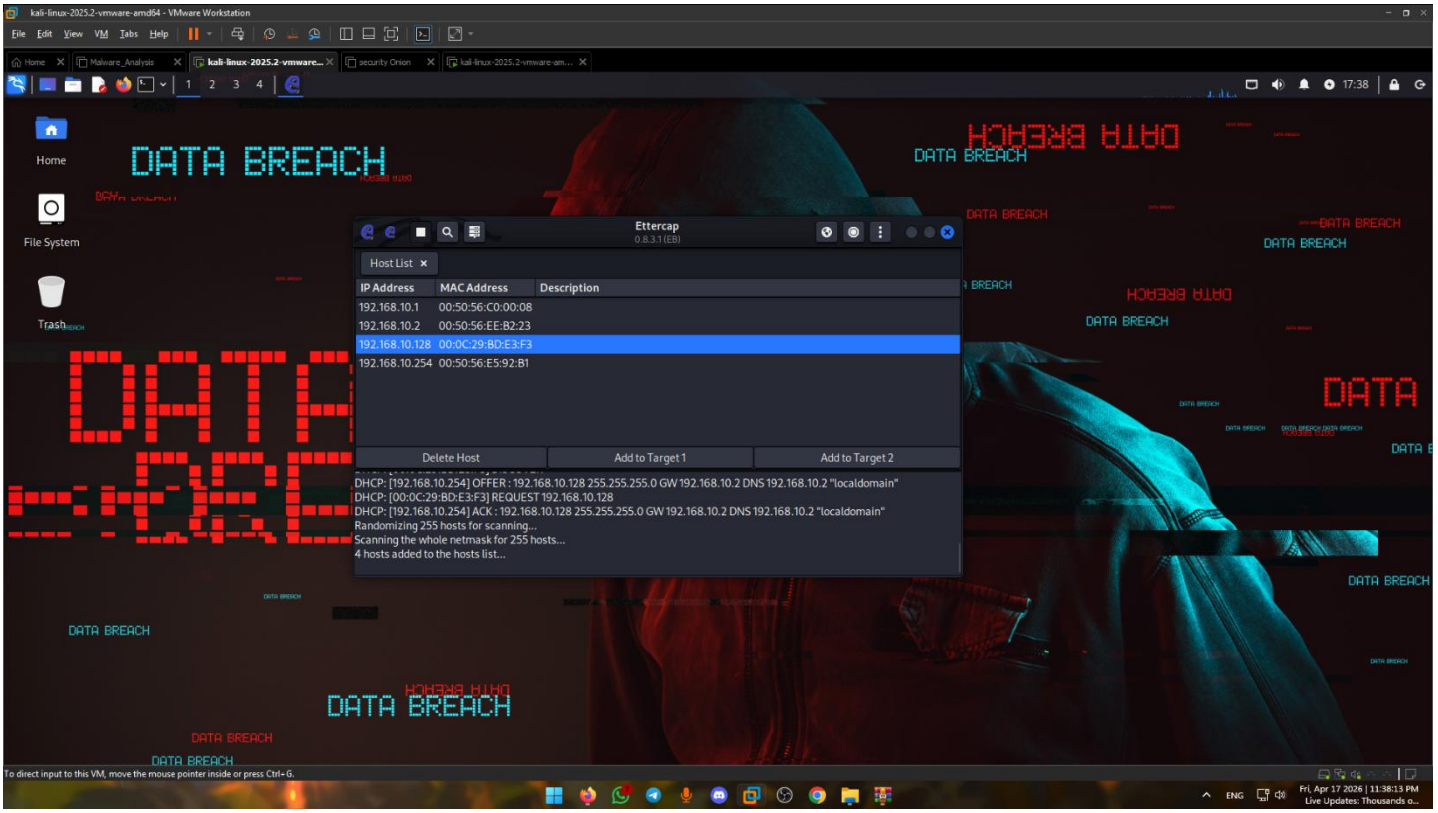
- Once the unified sniffing session started, the tool's menu was used to navigate to **Hosts > Scan for hosts** to discover active devices on the local network.
- Following the scan, the **Hosts > Hosts list** was opened. The scan successfully identified several IP addresses on the network, including:
    - 192.168.10.1
    - 192.168.10.2
    - 192.168.10.128
    - 192.168.10.254

Step 4: Target Selection

- From the Host List, specific machines were designated as targets for the interception.
- According to the Ettercap console logs at the bottom of the interface:
    - The host with IP 192.168.10.128 was added as **TARGET 1**.

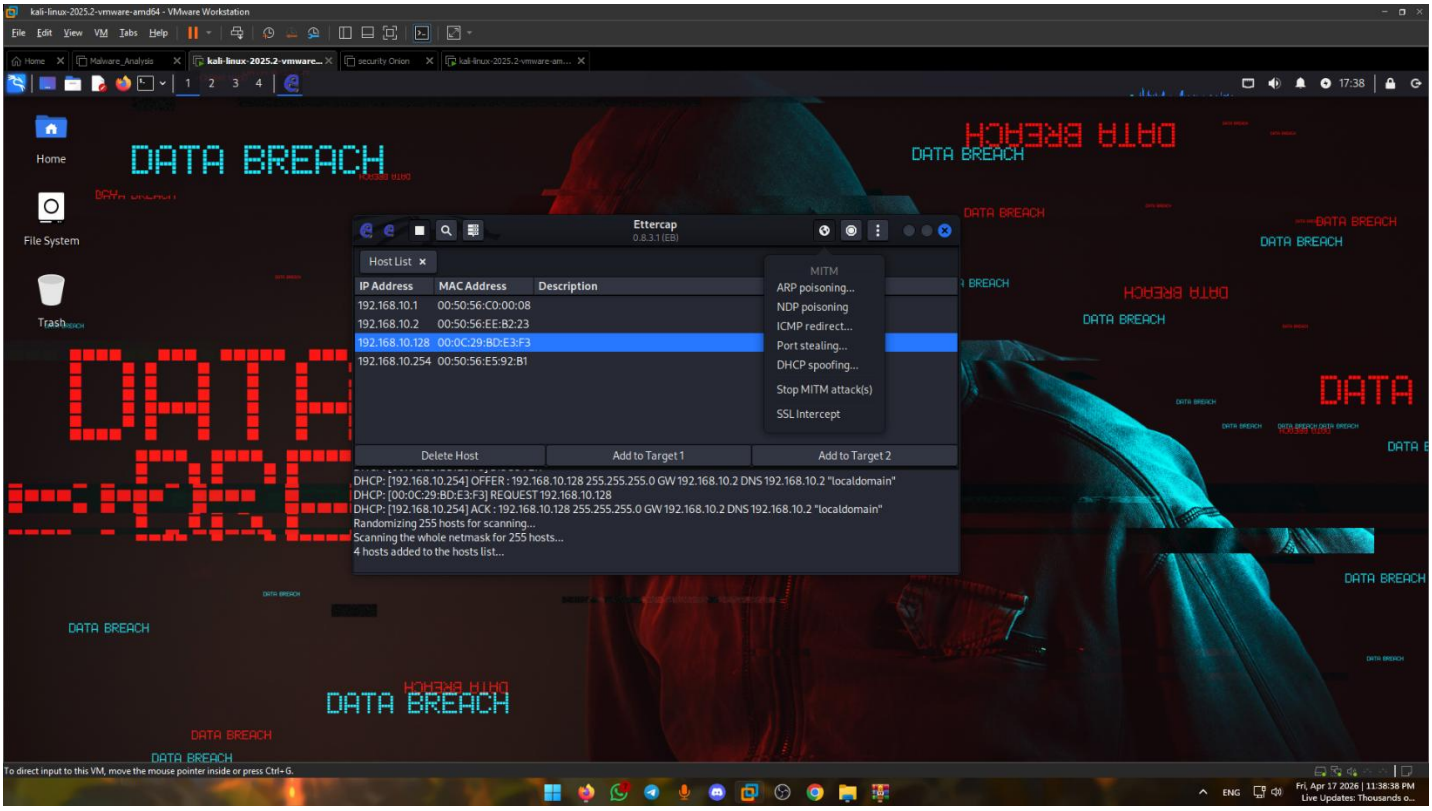
- The host with IP 192.168.10.2 was added as **TARGET 2**.

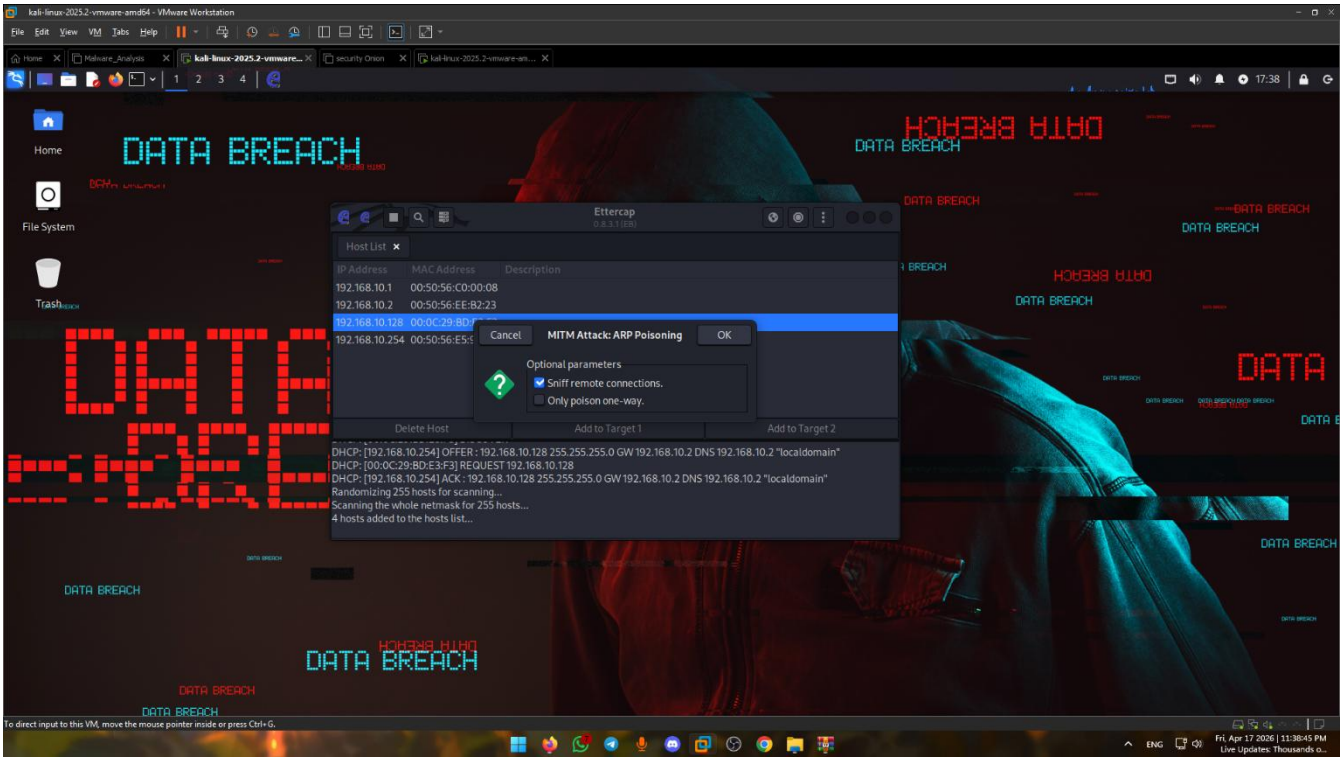


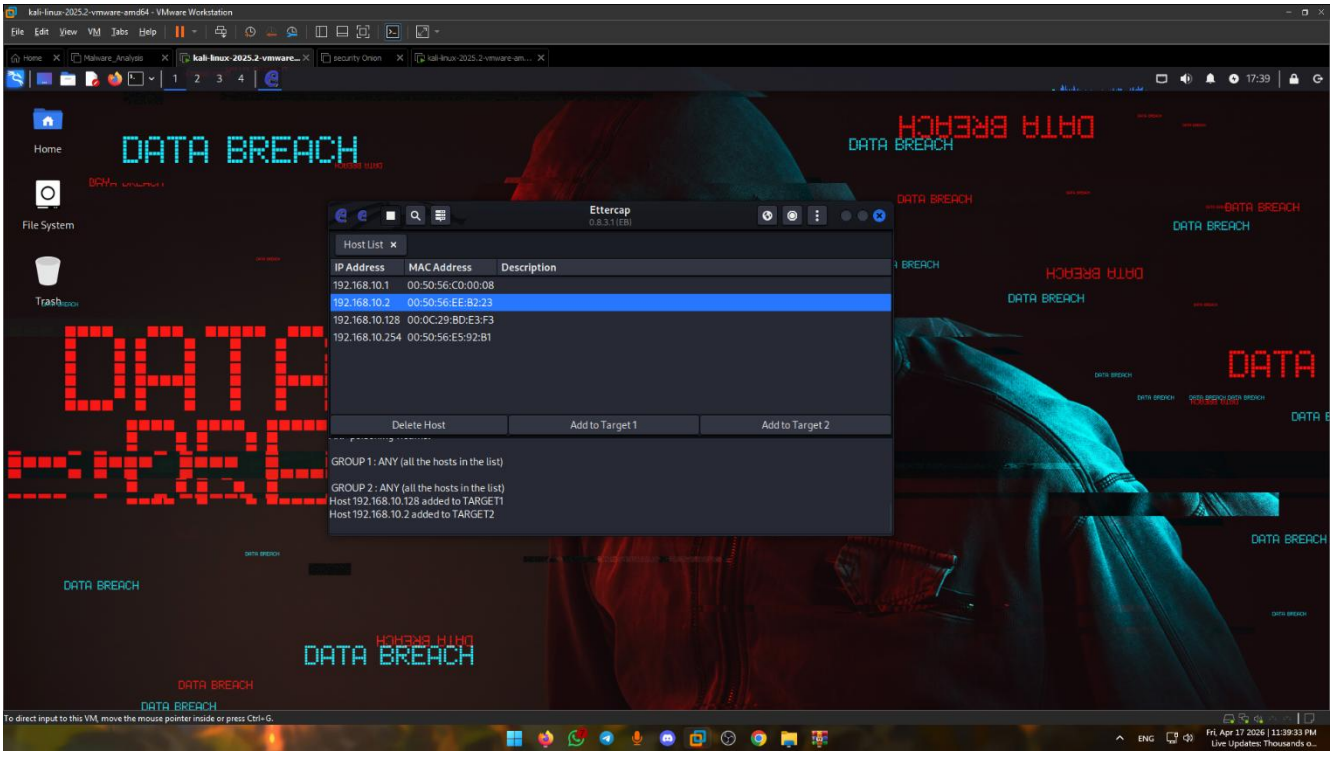


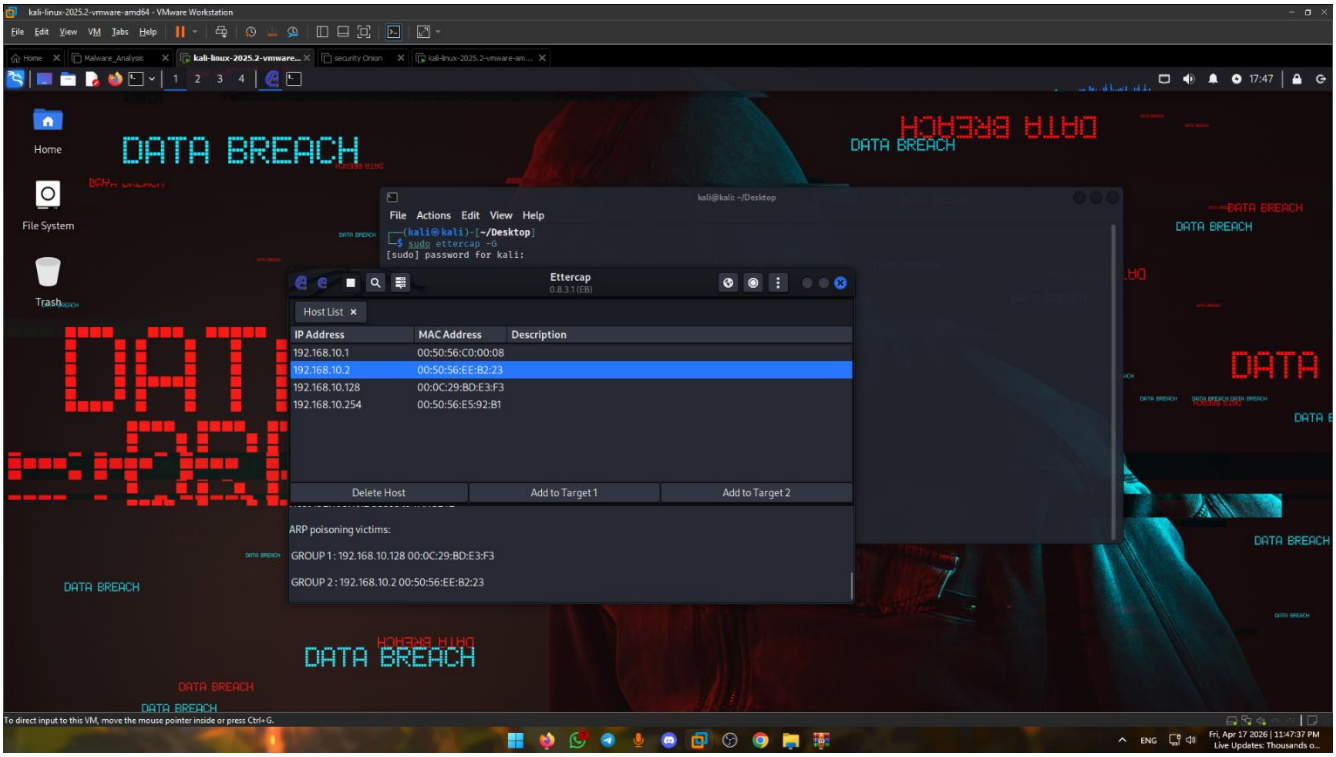
Step 5: Executing the MITM Attack

- With targets assigned, the MITM menu (represented by a globe icon) was accessed, and **ARP poisoning...** was selected.
- In the configuration prompt for the attack, the optional parameter **"Sniff remote connections"** was checked.
- The attack was initiated. The Ettercap console output confirmed the successful deployment, listing the ARP poisoning victims as Group 1 (192.168.10.128) and Group 2 (192.168.10.2).




kali-linux-2025.2-vmware-amd64 - VMware Workstation
Ettercap
Host List
IP Address
MAC Address
Description
192.168.10.1 00:50:56:C0:00:08
192.168.10.2 00:50:56:EE:B2:23
192.168.10.128 00:0C:29:BD:E3:F3
192.168.10.254 00:50:56:E5:92:B1
Cancel
MITM Attack: ARP Poisoning
OK
Optional parameters
Sniff remote connections.
Only poison one-way.
Delete Host
Add to Target 1
Add to Target 2
DHCP: [192.168.10.254] OFFER : 192.168.10.128 255.255.255.0 GW 192.168.10.2 DNS 192.168.10.2 "localdomain"
DHCP: [00:0C:29:BD:E3:F3] REQUEST 192.168.10.128
DHCP: [192.168.10.254] ACK : 192.168.10.128 255.255.255.0 GW 192.168.10.2 DNS 192.168.10.2 "localdomain"
Randomizing 255 hosts for scanning...
Scanning the whole netmask for 255 hosts...
4 hosts added to the hosts list...
DATA BREACH


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192.168.10.128 00:0C:29:BD:E3:F3
192.168.10.254 00:50:56:E5:92:B1
Delete Host
Add to Target 1
Add to Target 2
GROUP 1 : ANY (all the hosts in the list)
GROUP 2 : ANY (all the hosts in the list)
Host 192.168.10.128 added to TARGET1
Host 192.168.10.2 added to TARGET2
DATA BREACH



Conclusion

The lab successfully demonstrated the procedural steps to initialize Ettercap, scan a local subnet for live hosts, assign targets, and execute an ARP poisoning attack to potentially intercept traffic between the specified IP addresses.